able to exploit the fastest competing consolidator or self-aggregator options available on behalf of their retail clients. At this level of market data access, the Commission believes that that market participants will make decisions about what sort of competing consolidator product to use (or whether to self-aggregate) based off of competitive business considerations, that the final rules will make the options available cheaper than today, and that this will all work to the benefit of retail investors when these market participants work on their behalf to execute orders.

(c) Effects from the Interaction with the Consolidated Audit Trail

(i) CAT Baseline

17 CFR 242.613 (Rule 613) of Regulation NMS requires the national securities exchanges and national securities associations ("SROs") to jointly develop and file with the Commission a national market system plan to create, implement and maintain a consolidated audit trail ("CAT").[2420] At the time of adoption, and even today, trading data was and is inconsistent across the self-regulatory organizations and certain market activity is difficult to compile because it is not aggregated in one, directly accessible consolidated audit trail system. The goal of Rule 613 was to require the SROs to create a system that provides regulators with more timely access to a sufficiently comprehensive set of trading data, enabling regulators to more efficiently and effectively reconstruct market events, monitor market behavior, and identify and investigate misconduct. Rule 613 thus aims to modernize a reporting infrastructure to oversee the trading activity generated across numerous markets in today's national market system.

[2420] See supra note 1220.

On November 15, 2016, the Commission approved the national market system plan required by Rule 613 ("CAT NMS Plan" or "Plan") that was filed by the self-regulatory organizations.[2421] In the CAT NMS Plan, the SROs described the numerous elements they proposed to include in the CAT, including, (1) requirements for the plan processor responsible for building, operating and maintaining the Central Repository,[2422] (2) requirements for the creation and functioning of the Central Repository, (3) requirements applicable to the reporting of CAT Data by SROs and their members. "CAT Data" is defined in the CAT NMS Plan as "data derived from Participant Data, Industry Member Data, SIP Data, and such other data as the Operating Committee may designate as 'CAT Data' from time to time."[2423]

The CAT NMS Plan requires SROs and their members to record and report various data regarding orders by 8:00 am the day following an order event.[2424] The Plan requires industry members to record timestamps for order events in millisecond or finer increments with a clock synchronization standard of within 50 milliseconds.[2425] The CAT NMS Plan Processor, FINRA CAT, is then required to process the order data into a uniform format, link the entire lifecycle of each order, and combine it with other CAT Data such as SIP Data.[2426] The Plan Processor is also required to store CAT Data to allow the ability to return results of queries on the status of order

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2421 See id.

2422 The Central Repository is the repository responsible for the receipt, consolidation, and retention of all information reported to the CAT. See CAT NMS Plan, supra note 1220, at Section 1.1.

2423 See id. The Operating Committee is the governing body of the CAT NMS Plan.

2424 See id. at Sections 6.3 and 6.4.

2425 See id. at Section 6.8.

2426 See id. at Section 6.5.

books at varying time intervals.[2427] Regulators, such as the Commission and SROs will use the resulting CAT Data only for regulatory purposes such as reconstructing market events, monitoring market behavior, and identifying and investigating misconduct.[2428] At this time, the Commission has little information about what specific data, in addition to CAT Data, such as proprietary depth of book and auction data, the SROs currently intend to include in their enhanced surveillance systems.[2429]

(ii) Economic Effects on CAT

The Commission recognizes that the final rules could affect the Consolidated Audit Trail, resulting in benefits to investors from improved regulatory oversight, costs to CAT from potentially switching from a current SIP to a competing consolidator, costs to CAT from integrating consolidated market data into the CAT Data model, and costs to SROs of updating their enhanced surveillance systems to use consolidated market data provided by the CAT.[2430] Specifically, the Plan Processor for the Consolidated Audit Trail, FINRA CAT, is required to incorporate all data from SIPs or pursuant to an NMS plan into the Consolidated Audit Trail. If the Commission were to approve these amendments, the CAT NMS Plan Operating Committee could choose to purchase such data from a different entity and would be required to purchase the expanded consolidated data.

The Commission believes that the incorporation of the expanded data into CAT will improve regulatory oversight to the benefit of investors. As explained in the Approval order for

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2427 See id. at Section 6.5(c)(ii).

2428 See id. at Section 6.5(g); CAT NMS Plan Approval Order, supra note 1220, at 84833–34.

2429 See 17 CFR 242.613(f) (Rule 613(f)) of Regulation NMS.

2430 See supra Section IV.B.5 for a more detailed discussion of how the proposal would alter the requirements of the Consolidated Audit Trail NMS Plan.

the Consolidated Audit Trail, the expected benefits of the CAT include "improvements in regulatory activities such as the analysis and reconstruction of market events, in addition to market analysis and research…, as well as market surveillance, examinations, investigations, and other enforcement functions," and derive from improvements in four data qualities: accuracy, completeness, accessibility, and timeliness.[2431] Accuracy refers to whether the data about a particular order or trade is correct and reliable. Completeness refers to whether a data source represents all market activity of interest to regulators, and whether the data is sufficiently detailed to provide the information regulators require. Accessibility refers to how the data is stored, how practical it is to assemble, aggregate, and process the data, and whether all appropriate regulators could acquire the data they need. Timeliness refers to when the data is available to regulators and how long it would take to process before it could be used for regulatory analysis.

The Commission believes that the expanded consolidated data from the final rules could improve the completeness and accessibility of CAT Data.[2432] In particular, the final rules will improve the completeness of CAT Data because CAT Data would contain quotes smaller than

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2431 See CAT NMS Plan Approval Order, supra note 1220, at 84802–803.

2432 The Commission believes the final rules will not affect the accuracy or timeliness of CAT Data. The Commission does not believe that the proposal would alter the accuracy of timestamps of trades and quotes. While some competing consolidators might offer data that more accurately represents the data observed by certain market participants at the time of an order event, the Commission does not expect that all market participants would observe the exact same data at that order event, much like the case today. In addition, industry member clock synchronization and timestamps on the order events in CAT Data are not fine enough for the latency improvements to affect the accuracy of assigning an order event to the consolidated market data likely observed at the time of the order event. Finally, the order data in CAT is not required to be reported until 8:00 am the day following an order event. Hence, because latency improvements from the proposal would be measured in microseconds, the Commission does not believe that the final rules will improve the timeliness of CAT Data.

100 shares, depth of book information, and auction information. While the CAT will contain query functionality capable of recreating limit order books, the depth of book information will allow regulators to see the displayed order books that others see around the time of the order events. While the Commission does not know if SROs plan to incorporate depth of book and auction information into their enhanced surveillance systems or other regulatory activities using CAT Data, the proposal will improve the accessibility of consolidated market data for SRO and Commission CAT-related uses because SROs would have access to such data in a standardized format through the Consolidated Audit Trail instead of through the variety of formats currently used in proprietary data. The final rules will also improve accessibility because the SROs and Commission would have such data on the same system as CAT Data.

The Commission believes that the improvements in completeness and accessibility would facilitate more efficient regulatory activities using CAT Data that will benefit investors. In particular, the final rules could make broad-based market reconstructions using CAT Data more efficient by increasing the depth of information that could be incorporated into such reconstructions with current CAT Data. The Commission believes that depth of book information, quote information in sizes less than 100 shares, and auction information are all valuable in a broad-based market reconstruction. Further, the improvements would allow for more targeted surveillances and risk-based examinations using current CAT Data. For example, the depth of book information will be valuable when building surveillances to detect spoofing or in investigating spoofing because spoofing often involves creating a false impression of depth at prices outside of the best bid or offer. In addition, the auction information will facilitate auction market reconstruction to evaluate manipulation concerns and inform policy. Quote information in sizes less than 100 shares will facilitate analysis by regulators of broker-dealers' best

execution practices by providing potential execution prices that are better than the current NBBO in stocks priced over $250.[2433]

The Commission recognizes that the interaction between the final rules and the Consolidated Audit Trail could also create additional costs. Such additional costs are likely to be borne by SROs and their members. These costs could include switching costs, additional data costs, and data storage and processing costs. The proposal will result in switching costs if the Central Repository has to obtain the data from a different source. The source of the switching costs could be from changing data input formats and technical specifications, which would require one-time implementation costs. The Commission recognizes that the SIP technical specifications change a few times a year such that the switching costs associated with the proposal would be the costs in excess of the regular costs incurred when the SIP technical specifications change.[2434] The Commission at this time, cannot judge whether switching data providers would result in higher or lower on-going data intake costs but data intake costs presumably could be factored into the selection of a competing consolidator. Also, in order to continue to receive certain quotation and transaction data for OTC equities currently included in the SIP feeds, CAT would have to obtain such data from a different source, and would have to incur any associated costs in doing so.[2435] The Commission recognizes that increasing the amount of data managed and analyzed by CAT will increase the costs of data storage and

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2433 See supra Section V.C.1(b)(i) for data showing that odd-lot quotes in higher priced securities often improve upon the current NBBO.

2434 See CTA, Technical Documents, available at https://www.ctaplan.com/tech-specs (last accessed Jan. 30, 2020) (showing the SIP tech specs version history, which identifies the changes over the years); UTP Data Feed Services Specification, available at http://www.utpplan.com/DOC/UtpBinaryOutputSpec.pdf (showing the SIP tech specs version history, which identifies the changes over the years).

2435 See supra Sections II.C.2(c) and V.C.1(a) for a discussion of these potential costs.

processing to integrate the expanded data with other CAT Data. However, the Commission does not expect the final rules to substantially increase the costs of operating the CAT because any marginal increase in cost associated with consolidated market data will be dwarfed by the processing costs already incurred by CAT, which includes processing for all options quotation activity among other order lifecycle events and is significantly larger in size than consolidated market data.

The Commission recognizes that the final rules will result in SROs incurring costs to integrate additional CAT Data into their surveillances. Even if the SROs would otherwise include depth of book and auction information in the CAT-related surveillances, they would incur costs in changing their surveillances to use the data in CAT rather than using data from proprietary feeds.

The Commission also considered whether the requirements in CAT will impose costs as a result of CAT's effect on the competition among competing consolidators. Because the Commission does not believe CAT will significantly affect the competition among competing consolidators,[2436] it will not impose additional costs resulting from this effect.

The Commission believes that CAT implementation milestones will not be impacted by the final rules given that sufficient lead time will be available and integration efforts could be scheduled as part of standard release planning. The Commission believes that switching market data providers and expanding consolidated market data within CAT will require limited resources relative to the current implementation activities. Further, any resources devoted by SROs to updating their surveillances are separate from the efforts to implement CAT.

[2436] See infra Section V.D.2 for a discussion of the interaction between the proposal and CAT on competition among competing consolidators.

(d) Effects on Data Vendors

The Commission believes that the final amendments would have an effect on the broad financial data services industry. To the extent that the amendments lead to cheaper (relative to proprietary data feeds) and higher content consolidated market data products, the Commission expects that costs to data vendors would go down and the ability of such vendors to grow their customer base would increase. It is also possible that data vendors may increase the range and quality of products they offer using the new expanded core data and that new firms enter the data vendor business. To the extent that the risk of price increases for core data is realized instead, the Commission believes these businesses could potentially face higher costs, which when passed on to clients could cause their customer base to shrink. In the event that these outcomes are severe, it is possible that some data vendors could exit the market. The Commission is uncertain about the potential size and scope of these effects because it is unable to determine both the role of these costs in producing the products supplied by the data services industry and the extent to which the enhanced quality of new core data could play a role in the quality of their products.[2437]

[2437] One commenter stated that this information "should be essential" to the Commission's analysis. See Nasdaq Letter IV at 47. The Commission requested comment on the costs of market data vendors and the effect of new core data on their products and did not receive any. Data vendors are not required to disclose information to the Commission about the costs of their business at a level of detail sufficient to improve the Commission's understanding beyond what is said here. The assertions the Commission does make in this section about the effect of the rule on market data vendors do not depend on this information.

D. Impact on Efficiency, Competition, and Capital Formation

1. Efficiency

The Commission believes that the adopted amendments will have a number of different effects on efficiency. In particular, the Commission believes that the amendments will lead to more efficient gains from trade, improve the efficiency of order execution for some market participants, improve price efficiency, and affect how efficiently core data is distributed. The rest of this section discusses these different effects of the amendments on efficiency.

As discussed above, the Commission believes that the expansion of core data under the final amendments would increase transparency for market participants who do not currently access proprietary DOB feeds and allow them to more easily find liquidity that they can trade against.[2438] Currently, some of these market participants may not trade because they cannot see the quotes available to them, either through a lack of information about odd-lots, depth of book, or auction information. The Commission believes that the final amendments will alleviate some of this information shortage and will allow traders to more easily find counterparties. This may result in more voluntary trades occurring between market participants, which could lead to more efficient gains from trade, since these are trades which currently do not take place only because of a lack of information.[2439] However, if the inclusion of additional odd-lot, depth of book, or auction information does not induce additional voluntary trading from market participants who do not currently access proprietary DOB feeds, then the final amendments may not produce more efficient gains from trade.[2440]

2438 See supra Sections V.C.1(b), V.C.1(c).

2439 Id.

2440 Id.

The Commission believes that the expansion of core data could also improve the efficiency with which some market participants, or their broker-dealers, execute orders. As discussed above, by adding odd-lot, depth of book, and auction information to core data, the final amendments will reduce information asymmetry between broker-dealers and other market participants who subscribe to proprietary data feeds and users of current SIP data. This could improve the ability of broker-dealers and other market participants who currently do not have access to this information to trade against those market participants who do. As a result, this could improve the efficiency with which they execute their orders by allowing them to select a better trading venue or method of executing their order. Furthermore, for market participants who currently rely on exclusive SIPs for their order executions, the reduction in latency provided by the decentralized consolidation model could reduce the risk that their orders are picked off, which could reduce their adverse selection costs. This could potentially reduce their transaction costs and allow them to more efficiently achieve their investment or trading objectives or those of their clients.[2441]

As discussed previously, the Commission believes that there is some potential for new broker-dealers to become competitive in the market for sophisticated order execution as a result of this rule because they may be able to use the expanded content and lower latency of core data to develop SORs or other tools that allow them to compete more effectively with broker-dealers who currently base order execution decisions off of proprietary DOB data.[2442] To the extent that this happens, the clients of these broker-dealers could see their orders executed more efficiently and their execution costs reduced.

2441 Id.

2442 See supra Section V.C.4(b).